

An Exact Two-Radius Classification on Every L^p for Horizontal Spherical Means on H -Type Groups

Candidate full result for arXiv:2109.14963

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Status. Candidate full result, likely valid, pending specialist review. It gives sharp exceptional sets for every $1 \leq p \leq \infty$ and answers the first question in source Remark 4.1.

1 Source problem

E. K. Narayanan, P. K. Sanjay, and K. T. Yasser ask in Remark 4.1 of [1] whether a two-radius theorem holds above their one-radius injectivity threshold $2m/(m-1)$, including $p = \infty$.

rf. Now the proof can be completed as earlier. We omit the details. This completes the proof of Theorem [1.3](#)

Remark 4.1. The Abel summability result for the spectral decomposition will be true for all $1 < p < \infty$, if we can estimate the operator norm of $\mathcal{A}_k$. It is a natural question whether a two radius theorem is true for functions in $L^p(G)$ for $\frac{2m}{m-1} < p \leq \infty$ and whether our results can be proved for averages over K -orbits where $(G \rtimes K, K)$ is a Gelfand pair as in the case of the Heisenberg group. We hope to return to these questions and some others in the near future.

Remark 4.2. When $1 \leq p \leq 2$, it is possible to take the Fourier transform in the central variable and prove the injectivity results for the spherical means with weaker conditions of growth on the function. See [\[DS21\]](#)

The complete remark appears on physical page 19 of the arXiv PDF. The part settled here is:

“It is a natural question whether a two radius theorem is true for functions in $L^p(G)$ for $\frac{2m}{m-1} < p \leq \infty$.”

2 Definitions and exceptional sets

Let G be an H -type group whose Lie algebra splits orthogonally as $\mathfrak{v} \oplus \mathfrak{z}$, with $\dim \mathfrak{v} = 2n$ and $\dim \mathfrak{z} = m \geq 2$. Write points as $(z, t) \in \mathbb{R}^{2n} \times \mathbb{R}^m$. For $u > 0$, let μ_u be normalized surface measure on $\{(z, 0) : |z| = u\}$.

For $k \geq 1$, write the positive zeros of L_k^{n-1} as $0 < \zeta_{k,1} < \cdots < \zeta_{k,k}$, and set

$$\mathcal{E}_n = \left\{ \sqrt{\frac{\zeta_{k,i}}{\zeta_{k,j}}} : k \geq 1, 1 \leq i, j \leq k \right\}.$$

Let $0 < \beta_1 < \beta_2 < \cdots$ be the positive zeros of the ordinary Bessel function J_{n-1} , and set

$$\mathcal{B}_n = \left\{ \frac{\beta_i}{\beta_j} : i, j \geq 1 \right\}.$$

Both sets are countable, reciprocal-invariant subsets of $(0, \infty)$.

3 Proof Intuition

The bounded spherical spectrum has two branches. At nonzero central frequency, the multiplier of μ_u is a nonvanishing factor times $L_k^{n-1}(\lambda u^2/2)$; these are the Laguerre branches. At zero central frequency, the group becomes its Euclidean horizontal quotient and the multiplier is the normalized Bessel function $\mathcal{J}_{n-1}(\tau u)$. Thus two radii have a common spectral zero precisely on $\mathcal{E}_n \cup \mathcal{B}_n$.

For finite p , the zero-central-frequency branch cannot support a nonzero L^p function. Level-by-level support containment and Abel reconstruction therefore leave only $\mathcal{E}_n$. For $p = \infty$, Abel reconstruction is unavailable and the Bessel branch is real. The endpoint is instead closed by a Wiener–Tauberian theorem for the biradial L^1 algebra: absence of a common zero forces the annihilator ideal to be all of $L^1(G)$.

4 Exact all- p classification

Theorem 1. *Let G , m , n , and μ_u be as above, and let $r, s > 0$.*

1. *If $1 \leq p \leq 2m/(m-1)$, one radius is injective on $L^p(G)$.*
2. *If $2m/(m-1) < p < \infty$, the pair (μ_r, μ_s) is injective on $L^p(G)$ if and only if $r/s \notin \mathcal{E}_n$.*
3. *On $L^\infty(G)$, the pair (μ_r, μ_s) is injective if and only if $r/s \notin \mathcal{E}_n \cup \mathcal{B}_n$.*

Hence the horizontal two-radius problem is completely classified for every $1 \leq p \leq \infty$.

4.1 Finite- p classification

Part (1) is source Theorem 1.1 [1]. We recall the short argument for the new high finite- p range. The source spectral projections $\mathcal{A}_k$ are bounded on L^p for $1 < p < \infty$, and for $2 \leq p < \infty$ their Abel sums satisfy

$$f = \lim_{\rho \uparrow 1} \sum_{k=0}^{\infty} \rho^k \mathcal{A}_k f \quad \text{in } L^p(G). \quad (1)$$

Before using its low- p Fourier-support theorem, the proof of source Theorem 1.1 shows that $f * \mu_u = 0$ forces

$$\text{supp } \mathcal{F}_t(\mathcal{A}_k f(z, \cdot)) \subseteq \{a : L_k^{n-1}(|a|u^2/2) = 0\}$$

for almost every z . All steps through this containment use only the L^p projection bounds and (1); the upper bound on p enters only in the following sphere-support theorem.

If both radii annihilate f , a common support point at level $k \geq 1$ would give, for some i, j ,

$$|a|r^2/2 = \zeta_{k,i}, \quad |a|s^2/2 = \zeta_{k,j},$$

and hence $r/s = \sqrt{\zeta_{k,i}/\zeta_{k,j}}$. For $r/s \notin \mathcal{E}_n$ every support is empty; also $L_0^{n-1} = 1$. Thus $\mathcal{A}_k f = 0$ for all k , and (1) gives $f = 0$.

Conversely, at a ratio in $\mathcal{E}_n$, choose k, i, j and $\lambda = 2\zeta_{k,i}/r^2 = 2\zeta_{k,j}/s^2$. The source constructs

$$F_{k,\lambda}(z, t) = \frac{J_{m/2-1}(\lambda|t|)}{(\lambda|t|)^{m/2-1}} \varphi_k^\lambda(z)$$

up to normalization, with $F_{k,\lambda} * \mu_u = c_{k,n} \varphi_k^\lambda(u) F_{k,\lambda}$. It lies in L^p exactly for $p > 2m/(m-1)$, including infinity, and both eigenvalues vanish. This proves (2) and one endpoint obstruction.

4.2 The bounded endpoint

Let $R = L^1(G)^\natural$ be the commutative Banach algebra of biradial integrable functions. In our notation, its Gelfand spectrum [2, pp. 5–6] consists of

$$\Phi_{\lambda,k}(z,t) = c_k e^{-\lambda|z|^2/4} L_k^{n-1}(\lambda|z|^2/2) \mathcal{J}_{m/2-1}(\lambda|t|), \quad \lambda > 0, k \geq 0,$$

and the zero-central-frequency branch

$$\Phi_{0,\tau}(z,t) = \mathcal{J}_{n-1}(\tau|z|), \quad \tau \geq 0.$$

Only their zeros matter; $c_k \neq 0$ and $\mathcal{J}_\alpha(0) = 1$.

We use the following H-type Wiener–Tauberian result.

Lemma 1 (Sundari, Proposition 4.3). *If I is a proper closed right ideal in $L^1(G)$, then some character Ψ of R satisfies $\widehat{h}(\Psi) = 0$ for every $h \in I \cap R$.*

Sundari also constructs a biradial Poisson kernel $P \in R$ whose Gelfand transform is nowhere zero [2, Proposition 4.4].

Let $f \in L^\infty(G)$ and assume $f * \mu_r = f * \mu_s = 0$. Define

$$I_f = \{h \in L^1(G) : f * h = 0\}.$$

Young’s inequality makes I_f closed, and associativity makes it a right ideal. For $u = r, s$, put $h_u = \mu_u * P$. Convolution by a biradial finite measure preserves R , and

$$f * h_u = (f * \mu_u) * P = 0,$$

so $h_u \in I_f \cap R$. The Gelfand transforms are

$$\begin{aligned} \widehat{h}_u(\lambda, k) &= \widehat{P}(\lambda, k) c_k e^{-\lambda u^2/4} L_k^{n-1}(\lambda u^2/2), \\ \widehat{h}_u(0, \tau) &= \widehat{P}(0, \tau) \mathcal{J}_{n-1}(\tau u). \end{aligned}$$

Assume $r/s \notin \mathcal{E}_n \cup \mathcal{B}_n$. On every Laguerre character, at least one of $\widehat{h}_r, \widehat{h}_s$ is nonzero by the definition of $\mathcal{E}_n$; the assertion is automatic at $k = 0$. On every Bessel character, at least one is nonzero by the definition of $\mathcal{B}_n$, including $\tau = 0$. Thus no character annihilates both members of $I_f \cap R$. Lemma 1 shows that I_f cannot be proper. Hence $I_f = L^1(G)$. Applying an L^1 approximate identity to f yields $f = 0$ almost everywhere.

It remains to prove necessity of the Bessel condition. If $r/s = \beta_i/\beta_j$, choose $\tau = \beta_i/r = \beta_j/s$. The bounded, center-independent function

$$H_\tau(z,t) = \mathcal{J}_{n-1}(\tau|z|)$$

is a Euclidean spherical function on the horizontal quotient and satisfies $H_\tau * \mu_u = \mathcal{J}_{n-1}(\tau u) H_\tau$. Both eigenvalues vanish, so the pair is not injective. Together with the Laguerre example this completes the proof of Theorem 1.

5 Novelty and human review

Two bounded primary-source search rounds used the exact source question and variants involving “two radius”, “ H -type”, L^∞ , Wiener–Tauberian, Laguerre, and Bessel terminology. They found the source, its 2024 publication, Sundari’s 2008 theorem, and adjacent Heisenberg, Gelfand-pair, and X-ray results, but no statement of this exact classification. Novelty confidence is moderate pending specialist review.

The primary audit point is the standard measure-multiplier step $\mu_u * P \in R$ and the application of Lemma 1. Reviewers should also check the translation of Sundari’s spectrum notation and re-audit the finite- p support containment. The separate source question about general K -orbits is not addressed.

References

- [1] E. K. Narayanan, P. K. Sanjay, and K. T. Yasser, *Injectivity of spherical means on H -type groups*, arXiv:2109.14963; J. Fourier Anal. Appl. **30** (2024), Article 33, doi:10.1007/s00041-024-10089-9.
- [2] M. Sundari, *Tangential convergence of bounded harmonic functions on generalized Siegel domains*, arXiv:0803.2088; J. Aust. Math. Soc. **85** (2008), 419–430, doi:10.1017/S1446788708000700.